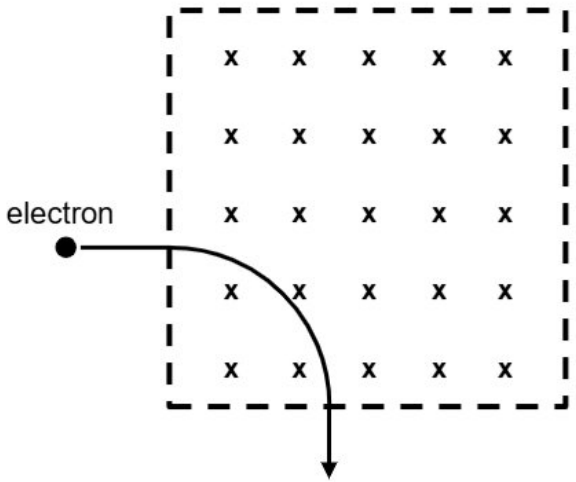
25 An electron with speed 3.0×10^6 m s^{-1} moves through a magnetic field of magnetic flux density 2.0 mT perpendicular to its direction of motion. The electron turns through an angle of 90° along a circular path as shown.



Which statement(s) is/are correct?

1 The force on the electron is 9.6×10^{-16} N.
2 The gain in energy of the electron is zero.
3 The change in momentum of the electron is zero.

A 1 only

B 1 and 2 only

C 1 and 3 only

D 3 only